



A dual-stage framework for cardiovascular abnormalities diagnosis from ECG signals using CA-GNN and semi-supervised autoencoders

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Abstract

Cardiovascular diseases (CVDs) cause global mortality, highlighting the need for effective strategies. Artificial intelligence (AI)-based applications have been used to diagnose myocardial infarction (MI) using automatic ECG signals. This study mainly introduces an effective MI diagnostic technique that incorporates advanced techniques. Signal processing, denoising, and analysis are performed sequentially to extract relevant features from electrocardiogram (ECG) signals. Clinicians can gain insight by understanding the patterns of these features. The relation of features is represented in a graph-based dataset. A novel compact attentional graph neural network (CA-GNN) model has been developed to classify cardiovascular diseases affected patients. The proposed model achieves an accuracy of 98.32% in the PTB dataset, indicating its robustness in diagnosing MI. However, to show the effectiveness and generalizability, the model was further experimented on two other datasets: MIT-BIH and PTB-XL; and achieved accuracies of 96.89%, 92.74%, respectively. This extension highlights the advantages of the proposed framework for diagnosing different types of cardiovascular disease abnormalities. In addition, this compact framework accelerates the training process, reduces the computation cost, and improves efficiency. This study also investigated a semisupervised autoencoder model to assess the performance of unlabeled data. The model obtains an accuracy of 89.43% on unlabeled PTB datasets and minimizes the loss to 0.095. The outcome of this study conveys an effective automated computer diagnostic system in cardiovascular classification abnormalities and paves the way for future endeavors to practice the classification of heart diseases from ECG signals.

Keywords Myocardial infarction · Signal processing · Graph neural network · Auto-encoder · Semi-supervised

1 Introduction

Myocardial infarction (MI) is one of the most common cardiovascular diseases and causes a high number of deaths annually [1]. During the past 30 years, since 1985, the suffering of MI has increased, with nearly 17 million people worldwide experiencing heart attacks [2]. MI is increasing

rapidly, and early and effective treatment of these cardiovascular abnormalities can reduce the risk of long-term health problems. AI has been widely used in the medical domain for its advantages in early diagnosis. This study also aims to facilitate AI in the framework to propose an effective early diagnosis method.

In recent years, several computer-aided methods have been developed to automatically detect and locate cardiovascular abnormalities. These methods often involve feature extraction, signal-to-image restoration, and classification. Feature extraction methods include time domain analyses [3], frequency domain analyses [4], optimization algorithms [5], the double recursive algorithm, and deep learning (DL) techniques [6, 7]. Several machine learning (ML) algorithms have been extensively used for MI, arrhythmia classification [8]. To reduce feature dimensions, Locality Preserving Projections (LPP) and Principal Component Analysis (PCA) have been used [9]. Additionally, DL architectures like Densely Connected Convolutional Networks, Generative Adversarial

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Networks (GAN), and transfer learning have been applied to physiological signals like the ECG [10]. Several studies [11–14] also developed hybrid models to classify cardiovascular diseases from ECG signals.

However, earlier studies on cardiovascular disease diagnosis have several significant limitations, primarily due to inadequate signal denoising and insufficiently detailed signal analysis. Thus, they fail to provide essential decision-making information. We address these gaps by integrating advanced denoising methods, extensive feature extraction, and a compact attentional graph-based model. Additionally, a semi-supervised technique is experimented with to assess the performance of the unlabeled data. Our framework demonstrates its effectiveness in automatically classifying multiple types of cardiovascular abnormalities using the ECG signals. The key contributions of this study are below:

- Several filtering techniques are used to reduce the noise and distortion from the signal.
- Design intuitive functions to detect the QRS complex, P waves, and T waves from the ECG signal.
- Multiple critical marker-based signal features, including amplitude, interval, and angle features, are extracted.
- Propose a compact threshold-based attentional graph model to classify the MI from ECG signals efficiently.
- Implement a semi-supervised auto-encoder model to test the performance of unlabeled data.

2 Related work

Several studies have been proposed over the years for effective classification of cardiovascular abnormalities from ECG signals. This section reviews state-of-the-art (SOTA) models and algorithms for MI classification.

Ma et al. [15] introduced a method combining shallow and deep features. Using artifact removal, waveform localization, and Hilbert curve transformation, their model achieved 98.95% accuracy on the PTB dataset, though no signal analysis was included. Jahmunah et al. [1] explored CNN and DenseNet for automated MI classification using pre-processed PTB ECG signals. Both models achieved over 95% accuracy. Pan et al. [16] proposed a multi-task channel attention network leveraging 12-lead ECGs on PTB and PTBXL and yielded over 90% accuracy. Gustafsson et al. [17] developed a DL model classifying control, NSTEMI, and STEMI, achieving C-statistics of 0.991 (STEMI) and 0.832 (NSTEMI). Han et al. [18] combined a DL model with a knowledge graph for MI location, severity prediction, and incorporated ontological structures and production rules, achieving 93.5% accuracy. Li et al. [19] addressed data imbalance using SLC-GAN to synthesize single-lead

ECGs, attaining 99.06% accuracy on the PTB Diagnostic ECG Database.

Traditional ML models are also widely employed for classification based on extracted ECG features. Moghadam et al. [11] extracted morphological features to diagnose MI and achieved 96% accuracy. However, they did not include denoising techniques, which could enhance feature extraction. Sraitih et al. [20] applied several ML models on the PTB and MIT-BIH datasets for MI prediction, focusing solely on traditional ML approaches without exploring advanced techniques or signal analysis. Barua et al. [21] proposed a hybrid model with multilevel feature extraction and iterative feature selection using 12-lead ECGs. The best lead-wise accuracy was 99.85%, with the majority voting accuracy being 99.94%.

Transformer-based models have shown excellent performance in time-series tasks [22], but require large datasets to generalize well [23]. As our study emphasizes relational data, we did not focus on transformer-based approaches. The most concerning limitation of the existing works is the lack of proper signal and feature correlation analysis. In this study, we aim to address these limitations (see Table 1 for comparison).

3 Methodology

Figure 1 displays the proposed methodology of this study. The subsequent sections elaborate on the methodological innovation of this study.

3.1 Dataset description

This study uses 3 datasets, including PTB, PTB-XL, and MIT-BIH. The PTB dataset was established by the Cardiovascular Department at the Benjamin Franklin University Clinic, Berlin [24] for automated MI classification research. PTB-XL [25] is the extended version of the PTB dataset, which has 5 classes: Normal, Myocardial, ST/T Change, Conduction Disturbance, and Hypertrophy. The MIT-BIH [26] dataset contains 5 classes where normal rhythms are indicated by N, V denotes ventricular beats, F refers to fusion beats, S is supraventricular beats, and Q is unknown beats. However, S has only two recordings that negatively impact model performance and are subsequently discarded for the experiments. We applied the synthetic minority oversampling technique to balance the dataset. Table 2 summarizes the datasets.

3.2 Signal analysis

ECG signal analysis is crucial to reveal important information [27]. Denoising is the first step, removing distortions and

Table 1 Benchmark comparison of the outcome of the existing literature

Paper	Year	Dataset	Signal Denoising	Signal Analysis	Feature Extraction	Attentional Graph-based Model	Signal to Image Restoration
Ma et al. [15]	2022	PTB, PTB-XL, MNIST and Fashion-MNIST	×	×	×	×	✓
Pan et al. [16]	2022	PTB, PTB-XL	✓	×	✓	×	×
Gustafsson et al. [17]	2022	Private Dataset, PTB-XL	×	×	×	×	×
Han et al. [18]	2022	Private	×	✓	✓	×	×
Li et al. [19]	2022	PTB	×	×	×	×	×
Moghadam et al. [11]	2023	PTB	✓	✓	✓	×	×
Sraiti et al. [20]	2022	PTB, MIT-BIH	×	×	×	×	×
Barua et al. [21]	2023	Physiobank Database	×	×	✓	×	×
Jahmunah et al. [1]	2022	PTB	✓	✓	✓	×	×
Proposed Study	2025	PTB, PTB-XL, MIT-BIH	✓	✓	✓	✓	✓

Bold text represents the result of our proposed framework

artifacts [28], since noisy signals hinder accurate identification of fiducial points and affect feature extraction.

3.2.1 Signal denoising

This study employs a 3rd-order Butterworth filter [29] with a 1–100 Hz bandpass and 1000 Hz sampling to remove low- and high-frequency noise, such as baseline wandering and motion artifacts, while preserving signal integrity. The Butterworth filter's smooth transition from passband to stopband ensures effective noise reduction. Additionally, a notch filter [30] at 50 Hz with a quality factor of 30 removes power-line interference by attenuating specific frequencies without affecting the rest of the signal. Filter parameters were selected based on dataset characteristics and optimized through rigorous experiments for clear ECG measurements. Figure 2 illustrates the output of the proposed denoising approach, where the filtered signal is less distorted compared to the original signal. This clean filtered signal also shows the fiducial points and the waves more clearly.

3.2.2 Fiducial points detection

Following noise eradication, fiducial points are identified to extract desired features. Fiducial points refer to waveforms' highest, beginning, and end points [31]. An ECG signal comprises the P wave, QRS complex, and T wave, representing electrical activity across heart regions. This work presents a technique to identify ECG fiducial points. The find peaks function detects the R-peak in the QRS complex by treating the ECG as a 1D array. A height threshold of 0.6 filters out local minima, while a distance of 100 ensures a minimum gap between peaks. After detecting the R peak, the Q and S waves are found within 20ms before and after it. The QRS onset and offset are determined 20ms before the Q peak and 20ms after the S peak, respectively. Figure 3 shows the detected QRS complex and corresponding peaks.

Detection of T and P waves depends on the QRS complex. A search window locates the T wave from 100ms to 500ms after the R-peak. The custom find peaks function identifies the highest peak in this range as the T wave, with prominence set to 0.5. Prominence measures a peak's height relative to its surroundings. For the P wave, the window spans from 200ms to 40ms before the R-peak. The highest peak in this range is the P wave, detected using a prominence of 0.3.

3.3 Feature extraction

The ECG signal of an MI shows distinct changes depending on the infarction's type and location. This study extracts features based on the wavelength and peaks of the QRS complex, as well as the P and T waves, focusing on time intervals and temporal segments. Interval features measure the

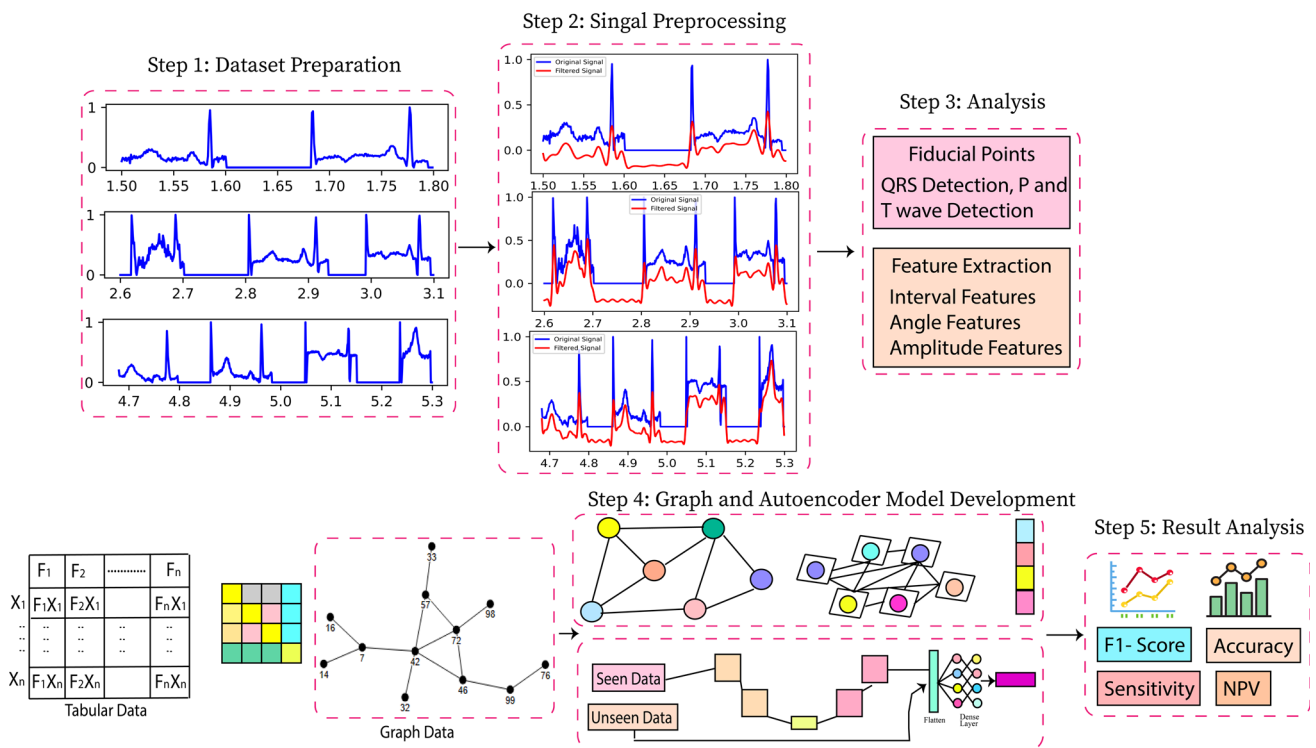


Fig. 1 The diagram of the proposed methodology

Table 2 Datasets overview of the proposed study

Param.	PTB	PTB-XL	MIT-BIH
Total Patients	290 subjects	18869 subjects	47 subjects
Lead System	15-lead (12 standard + 3 frank leads)	12-lead	2-channel
Number of Classes	2	5	4
Class Sample (Before Augmentation)	Normal-4047; Abnormal-10505	Normal ECG-9528; MI-5486; ST/T Change-5250; Conduction Disturbance-4907; Hypertrophy-2655	N-19966; V-680; F-351; Q-24
Class Sample (After Augmentation)	Normal-10505; Abnormal-10505	Normal ECG-9528; MI-9528; ST/T Change-9528; Conduction Disturbance-9528; Hypertrophy-9528	N-19966; V-19966; F-19966; Q-19966

Bold text represents the result of our proposed framework

time between waves, while segment features capture periods of electrical inactivity between waves. Amplitude features quantify ratios between the R-peak and other peaks: RPa, RQa, RSa, and RTa compare the R-peak with P-, Q-, S-, and T-peaks, respectively. Angle features are calculated relative to the upward direction and include angle Q, angle R, and angle S. Angle Q is the angular displacement around the Q wave; angle R is between the lines connecting Q and S waves to the R-peak; and angle S measures the angle change between lines linking the S wave to the T-peak and the R-peak.

The extracted signal features include **interval features** such as PR interval (PR_I), PR segment (PR_S), QT interval (QT), RT segment (RT), RS segment (RS), RQ segment (RQ), RP segment (RP), ST segment (ST_S), and ST interval (ST_I); **angle features** like Angle Q ($\angle Q$), Angle R ($\angle R$), and Angle S ($\angle S$); and **amplitude features** comprising RQ amplitude (RQ_A), RT amplitude (RT_A), RP amplitude (RP_A), and RS amplitude (RS_A).

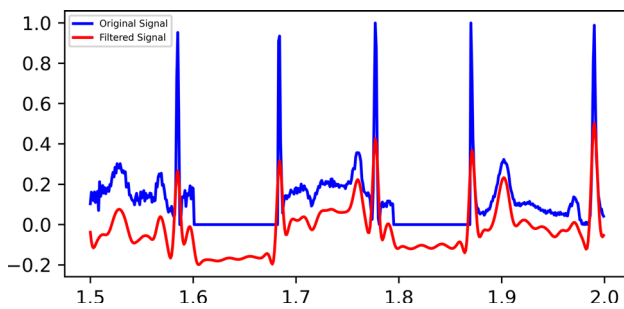


Fig. 2 Comparison of the original and the filtered signal

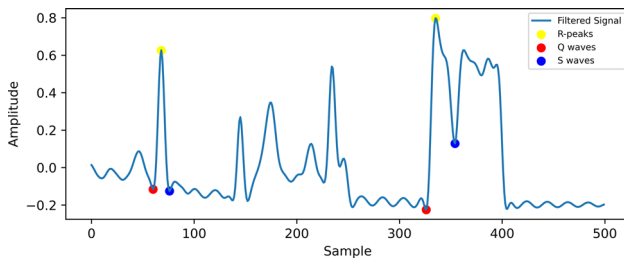


Fig. 3 The detection of the QRS complex and R, Q, and S peaks

3.4 Graph dataset construction

Converting tabular data into a graph space is challenging; we address this by combining cosine similarity with a KNN strategy. First, KNN retains the top k most similar nodes per sample, clustering related nodes. Then, cosine similarity evaluates pairwise similarity among these nodes. This KNN+cosine similarity approach ensures connectivity in high-dimensional spaces. The tabular dataset contains 8276 rows and 17 columns; each row ID corresponds to a graph node, and column values are node features. Edges are created with a similarity threshold of 0.90. The resulting undirected graphs have 8276 nodes and 28237 edges (PTB), 79864 nodes and 511569 edges (MIT-BIH), and 38112 nodes and 265245 edges (PTB-XL). Finally, this graph is used for MI classification via the proposed GCN model

3.5 CA-GNN model description

In recent years, Graph Convolution Networks (GCN) have been widely used to analyze graph-structured data for node classification [32]. In our framework, GCN classifies ECG samples using both individual features and their relational structure (i.e., KNN+cosine similarity constructed graph). Unlike traditional neural networks, GCNs aggregate information from neighboring nodes, leveraging correlated patterns in ECG features.

The core working mechanism of GCN comprises three steps: initialization, neighborhood convolution, and information aggregation. Each sample ID is a node with extracted

signal features as node attributes. Convolution enables message passing to predict node labels. To address drawbacks like redundant features, increasing noise, and complexity, we propose a two-step method: an adjacency learning module calculates attention scores and applies autonomous thresholding to select neighbors; then, an attentional graph operation aggregates attention weights to predict classes. The adjacency module computes absolute differences (i.e., cosine similarity) between node features, treated as attention scores. The masked attention mechanism [33] is applied to our graph-structured data, calculating the attention scores e_{ij}^A for all node j within the neighbor set $\mathcal{N}$ of node i . The equation (1) calculates the attention score of the corresponding node.

$$e_{ij}^A = |\phi(x_{[i,:]}W_A) - \phi(x_{[j,:]}W_A)|v_A, \quad j \in \mathcal{N}_i^C \quad (1)$$

Here, W_A is the attention weight matrix, x is the feature vector for node i to all other connected nodes, ϕ is a non-linear feature transformation, v_A is learnable attention vector, and $j \in \mathcal{N}_i^C$ is the set of neighborhood for node i . For each feature $x_{[i,j]}$, the attention weight matrix represents the arrangement of node features for that specific feature. A multi-layer perceptron layer is further utilized to determine how strong the edge is based on the attention score of a node within its neighbor nodes. To further modulate the absolute difference of the attention score between the node features and projecting into a scalar value, an attention vector v_A is utilized. Finally, the node-to-node relations are extracted. These attention scores are subsequently compared autonomously with the threshold using Equation (2):

$$T = \max(x_{[i,-1]}) \cdot \text{Sigmoid}(e_{ij}^A) \quad (2)$$

Here, max variable is the maximum attention score, the sigmoid function is utilized to normalize the attention score between 0 and 1, and T is the threshold matrix. The final neighbor nodes are estimated using this criterion, and the adjacency matrix (i.e., graph representation) is created. Following this, the convolution operation uses the value of this matrix for message passing and information aggregation. For this study, the proposed adjacency values of a node are determined using binary masking:

$$a_{ij} = \begin{cases} 1, & \text{if } x_{ij} < T \\ 0, & \text{otherwise} \end{cases} \quad (3)$$

Equation (2) determines the final neighborhood node, and Equation (3) indicates that if the feature vector x is less than the threshold T , then it will create the edges between nodes i and j ; otherwise, no edge will be formed. This technique dynamically prunes the connections between the nodes, leading to compact graph-structured input data.

After obtaining the final neighbor connection for each node, we implemented the multi-head graph attention layer to learn and aggregate the information. Equations (1), (2), and (3) help to determine optimal nodes, edges, and the overall connectivity, whereas the remaining equation is applied to the model layers. The layer numbers of the proposed model are manually tuned according to the dataset. The attention layer calculates the attention coefficients for each layer using Equation (4) and (5):

$$e_{ij}^{(k-1)} = \phi \left(\left[\widehat{h}_{[i,:]}^{(k-1)} \parallel \widehat{h}_{[j,:]}^{(k-1)} \right] W_{\text{att}}^{(k-1)} \right) v_{\text{att}}^{(k-1)} \quad (4)$$

$$\widehat{h}_{[i,:]}^{(k-1)} = h_{[i,:]}^{(k-1)} W^{(k-1)} \quad (5)$$

In Equation (4), v_{att} is the attention vector, W_{att} is the attention weight matrix of the k th layer. $\widehat{h}_{[i,:]}^{(k-1)}$ and $\widehat{h}_{[j,:]}^{(k-1)}$ are the feature vectors for two different nodes, i and j , ϕ is the non-linear activation function, and the resultant attention coefficient is stored in $e_{ij}^{(k-1)}$. Equation (5) is used to create the linear transformation of the features. It also helps to create a higher-level representation of the input features. Additionally, the attention coefficients need to be normalized for flexibility. Equations (6), (7) are applied to normalize the coefficients.

$$\alpha_{ij}^{(k-1)} = \frac{\exp(e_{ij}^{(k-1)})}{\sum_{j' \in \mathcal{N}_i^F} \exp(e_{ij'}^{(k-1)})} \quad (6)$$

$$h_{[i,j]}^{(k)} = \varphi \left(\sum_{j \in \mathcal{N}_i^F} \alpha_{ij}^{(k-1)} \widehat{h}_{[j,:]}^{(k-1)} \right) \quad (7)$$

After normalization, aggregation, and updated steps are sequentially executed. Here, $\alpha_{ij}^{(k-1)}$ is the normalized attention coefficient, and $\sum_{j \in \mathcal{N}_i^F}$ summation of aggregated message. The attention coefficient is weighted with the $(k-1)$ node. The activation function φ , ReLU, is applied to obtain the new node features, and the resulting updated hidden node is h . The final class prediction is the class label of this updated node. Our key novelty compared to traditional graph-based models is the advanced attention-based edge pruning, which

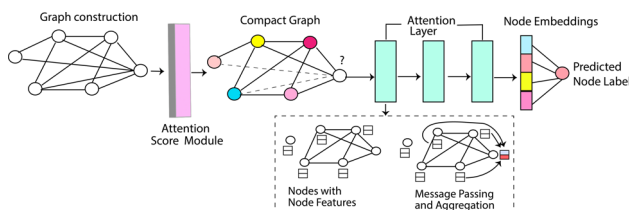


Fig. 4 The proposed graph dataset construction technique and CA-GNN model building

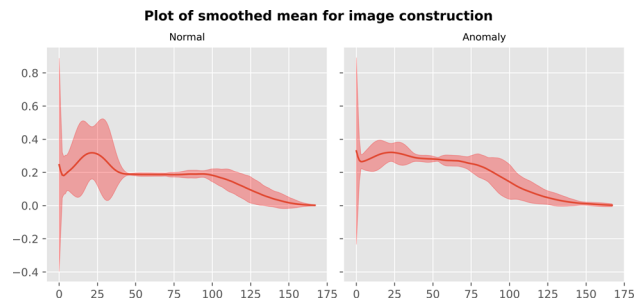


Fig. 5 Generated smoothed mean plot to feed into the auto-encoder model

makes it compact and highly adaptive. Moreover, it is deeply regularized with multi-head attention layers, which allows performing reliable classification tasks requiring less computational cost. Figure 4 displays the diagram of the proposed model.

3.6 Signal-to-image construction

To conduct an experiment using unlabeled data, this study aims to develop a semi-supervised model. This section describes signal-to-image reconstruction, which is mutually independent from the graph-based methods. The reconstructed images will be used as input for our semi-supervised model. The input shape and format are completely different from the input or data type of the CA-GNN model. This study introduces an intuitive algorithm to construct an image of the corresponding signal class. It first groups the target class, calculates the meaning for the grouped data, and converts it to a NumPy array. The standard deviation of this array is then calculated using an appropriate window size to smooth the data. Next, the confidence interval margins are determined (according to the data values). We create a subplot for each class and assign its respective label. Finally, the layout is adjusted to the figure to ensure the clarity of the images. This visualization technique effectively depicts the proper figure, which the auto-encoder model later utilizes for feature extraction. Figure 5 illustrates the generated plot for both normal and MI classes.

3.7 Proposed semi-supervised autoencoder model

We employ the semi-supervised autoencoder to assess how well the model can generalize to unlabeled data by learning a compact, lower-dimensional representation of the input features and then reconstructing them.

Figure 6 illustrates the proposed model, where the encoder generates a low-dimensional embedding output while the decoder reconstructs this embedding space using the up-sampling technique [34]. In between, a bottleneck is applied as a latent space. This compresses the encoded features,

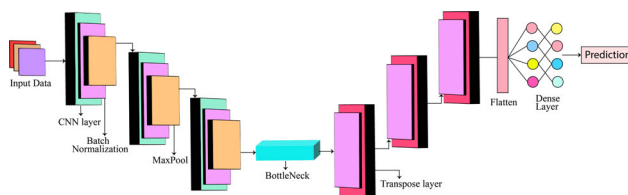


Fig. 6 Proposed auto-encoder model to test on unlabeled data

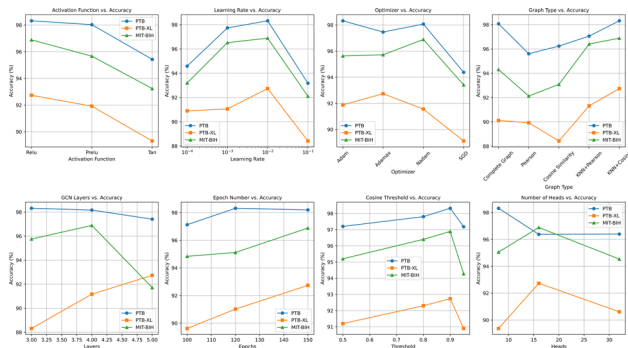


Fig. 7 The fine-tuning of different hyperparameters of the proposed model

reduces dimensionality, and creates an efficient representation. The encoder part comprises 3 CNN blocks, batch normalization, and a maxpool layer. The decoder mirrors the encoder's architecture with a transpose layer to perform the up-sampling operation. After the decoder, a flattened layer is added that converts the output of the decoder into a feature vector. A subsequent dense layer takes the input from the flattened layer, performs a matrix multiplication operation, and applies the activation function for prediction.

4 Results analysis

This section includes the experimental details of the proposed model.

4.1 Ablation study

This study investigated eight key hyperparameters to identify the optimal configuration of the model, as illustrated in Fig. 7.

Here, ReLU outperformed other activation functions in terms of accuracy for each dataset. The best learning rate was 0.01. To guide the model in training and minimize the loss, Adam performed better. The most crucial part of this ablation study is the selection of optimal graph construction techniques. Several graph construction techniques were investigated, including the complete graph, Pearson Correlation, Cosine Similarity, KNN+Pearson, and KNN+Cosine. The best result was obtained using the KNN+Cosine. The

Table 3 Performance analysis of the proposed model across three datasets

Metric	PTB	MIT-BIH	PTB-XL
Accuracy	98.32	96.89	92.74
Specificity	98.04	97.63	94.80
Recall	98.11	96.42	90.68
NPV	97.84	96.67	92.42
F1-score	97.65	96.88	90.96
FPR	0.72	0.237	1.52
FNR	1.14	0.711	1.358
FDR	1.1392	0.678	1.1392

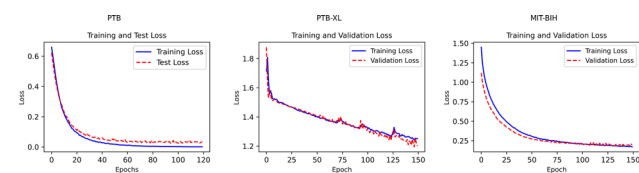


Fig. 8 Loss Curve of the proposed model for PTB, PTB-XL, and MIT-BIH datasets

key factor for superior performance is capturing local structures and keeping directional similarity between data points. Unlike complete graph or Pearson-based methods, KNN focuses on relevant neighbors, reducing noise and overfitting from weak connections. Cosine similarity strengthens this by assessing angles between feature vectors, ensuring robustness to scale variations. This combination creates a graph connecting nodes by proximity and orientation, reflecting true data relationships and enhancing model performance. The best performance for the PTB dataset is achieved with a configuration of 3 layers (120 epochs), whereas 4 and a maximum of 5 layers with 150 epochs result in the best performance for MIT-BIH and PTB-XL, respectively. The increase in layer number is observed with the increment in sample size.

4.2 Proposed graph model result analysis

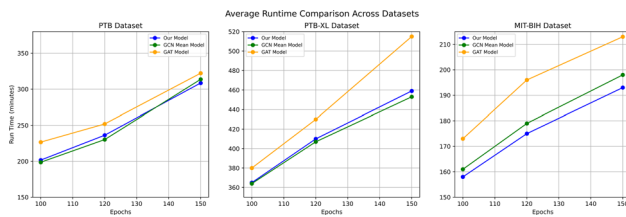
Table 3 presents the performance of the proposed model in the PTB, MIT-BIH, and PTB-XL datasets. It achieves accuracies of 98.32%, 96.89%, and 92.74%, respectively, with binary classification in PTB and multiclass on the others.

Figure 8 is utilized to display the loss curves of the proposed model. Each of the loss curves shows a steady decrease over the epoch and finally reaches the minimum loss score. Table 4 compares the proposed model with baseline methods. A key contribution is reduced training time by pruning redundant edges via attention-based dynamic edge filtering, creating a compact graph. Figure 9 shows that the proposed model consistently runs faster than GAT and GCN. Though

Table 4 Performance (in %) comparisons of the proposed CA-GNN model with existing SOTA models

Model	PTB			MIT-BIH			PTB-XL	
	Acc	Sen	Spe	Acc	Sen	Spe	Acc	Sen
1D CNN	86.37	85.5	87.2	84.85	83.8	85.9	78.22	77.0
SVM	89.24	88.6	89.8	81.67	80.6	82.7	81.73	80.8
DT	92.76	92.0	93.5	87.89	87.1	88.7	86.40	85.5
RF	95.83	95.4	96.2	89.05	88.3	89.8	89.95	89.2
XGB	96.27	95.9	96.6	89.03	88.3	89.7	85.82	85.24
GAT	97.81	97.4	98.68	96.30	95.28	96.79	92.14	91.5
GCN	97.85	97.5	98.29	92.52	91.7	93.3	85.48	84.73
Ours	98.32	98.11	98.04	96.89	96.45	97.63	92.67	92.36

Bold text represents the result of our proposed framework

**Fig. 9** Comparison of run-time evaluation with graph-based models**Table 5** Performance analysis of the proposed semi-supervised model

Metric	PTB	MIT-BIH	PTB-XL
Training Acc (%)	94.04	93.16	89.13
Test Acc (%)	89.43	92.75	81.83
Test Error	0.095	0.019	0.116

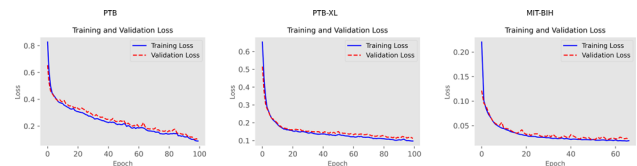
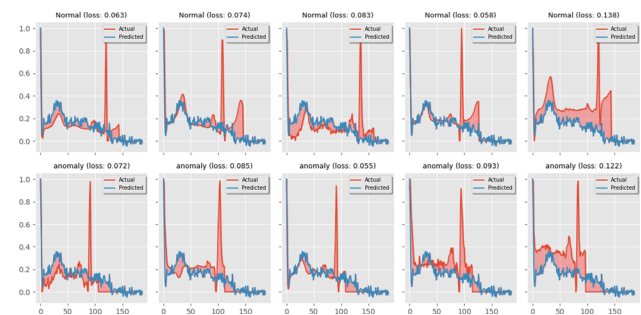
GCN is simpler and GAT is more complex, our optimized model achieves lower runtime with effective configurations.

4.3 Semi-supervised model result analysis

The proposed semi-supervised model achieved outstanding accuracy on unlabeled test data (see Table 5). We used 80% of the samples, and subsequently removed the label from the remaining data for testing.

The proposed model obtained high training accuracies of 94.04%, 89.13%, and 93.16% for PTB, PTB-XL, and MIT-BIH datasets, respectively. The highest accuracy was obtained in the MIT-BIH datasets. The test error for the unlabeled data is minimized at 0.095, 0.116, and 0.019 for the three datasets. Figure 10 displays the loss curve of the proposed auto-encoder model.

Figure 11 signifies the semi-supervised model outcome. In each scenario, the model returns the minimal signal loss. For the normal signals, the loss is between 0.063 and 0.138, and a 0.072 to 0.122 signal loss has been found for predicting abnormal signals.

**Fig. 10** Loss curves of the auto-encoder model highlighting the training convergence consistency**Fig. 11** Comparison of actual dataset signals vs. model predicted signals with loss scores

5 Discussion

The diagnosis of cardiovascular abnormalities has been widely studied, yet the transparency and depth of analysis behind predictions often limit reliability. This work presents a comprehensive method combining advanced techniques to classify cardiovascular-affected patients from ECG signals. The process begins with denoising ECG signals to remove artifacts that affect feature extraction. Subsequently, P, Q, R, and S peaks are detected, and signal-based features like amplitude and intervals are extracted. These features are essential markers of heart health conditions and provide valuable data for identifying distinct cardiovascular abnormalities. To explore feature relationships, we applied a graph-based approach using cosine similarity and KNN to connect nodes above a threshold. Experiment showed that on

both labeled and unlabeled data, the proposed model outperforms baselines.

Limitations include the need for larger or multi-modal datasets and additional denoising comparisons. Clinical validation was restricted due to data privacy and hospital access, but future work aims to collaborate with clinical institutions to address this.

6 Conclusion

This study focuses on diagnosing numerous life-threatening cardiovascular diseases. A comprehensive framework is proposed using ECG signals. To model the relationship between the features, CA-GNN is introduced, which yields a remarkable performance of 98.32% for binary classification in the PTB dataset, and reliable 96.89%, and 92.74% accuracy for multiclass classification in the MIT-BIH and PTB-XL datasets. Another significant contribution of this study is the development of a semi-supervised autoencoder model, which also achieves a reliable performance on unlabeled data. This research addresses the limitations of previous works and contributes to the advancement of cardiovascular abnormalities diagnostic systems based on ECG signals.

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Data Availability This study conducts all the experiments on 3 publicly available datasets: PTB [24], PTB-XL [25], and MIT-BIH [26].

Declarations

Conflict of Interests On behalf of all authors, the corresponding author states that there is no conflict of interest.

Ethics Approval and Consent to Participate Not Applicable.

Informed Consents Not Applicable.

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